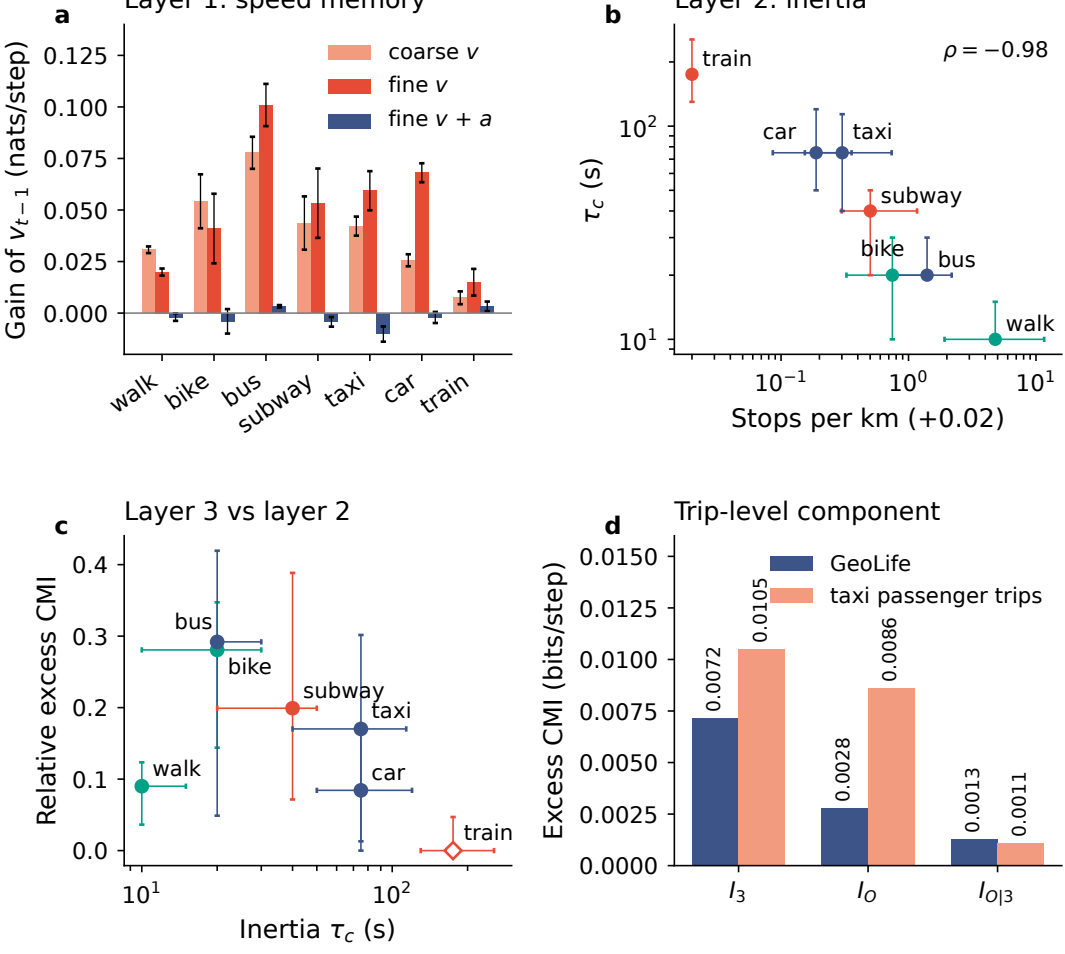
a
Layer 1: speed memory
coarse v
fine v
fine $v + a$
Gain of v_{t-1} (nats/step)
0.125
0.100
0.075
0.050
0.025
0.000
walk
bike
bus
subway
taxi
car
train
b
Layer 2: inertia
$\rho = -0.98$
train
car
taxi
subway
bike
bus
walk
τ_c (s)
10^2
10^1
10^{-1}
10^0
10^1
Stops per km (+0.02)
c
Layer 3 vs layer 2
Relative excess CMI
0.4
0.3
0.2
0.1
0.0
bus
bike
subway
taxi
car
walk
train
10^1
10^2
Inertia τ_c (s)
d
Trip-level component
Excess CMI (bits/step)
0.0150
0.0125
0.0100
0.0075
0.0050
0.0025
0.0000
GeoLife
taxi passenger trips
0.0072
0.0105
0.0028
0.0086
0.0013
0.0011
I_3
I_O
$I_{O|3}$
(b, c) right of way
non-motorised
shared road
own track